

HireFlow Agent

Agentic AI Recruitment Assistant powered by LangGraph.

LANGGRAPH • RAG • HUMAN-IN-THE-LOOP

Team Vcoders



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Github Link

<https://github.com/SyedUzair/hirflow>

The Recruiter's Bottleneck

Manual Overload

Screening 100s of resumes manually for a single Job Description (JD) leads to severe cognitive fatigue and missed top-tier talent. Keyword matching is outdated and fails to capture semantic meaning.

Fragmented Workflows

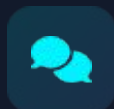
Candidate comparison is painfully slow. Generating targeted interview questions and conducting accurate, live salary research across disparate platforms shatters operational efficiency.

The AI Recruiter Co-Pilot



Analyze & Rank

Instantly parse complex JDs into structured data. Execute semantic searches across resumes to automatically rank candidates with high precision.



Generate & Research

Automatically generate candidate-specific interview questions and utilize tools (Tavily) to fetch live, real-world salary data and market expectations.



Human Finality

AI recommends. Humans decide. The system handles data synthesis, but relies on explicit human approval for shortlisting and final hiring decisions.

Multi-Node Agent Workflow



State-Based Routing via LangGraph



Centralized State Memory

The LangGraph State object acts as the brain. It seamlessly stores conversation history, parsed JD information, candidate ranking results, budget constraints, and current shortlists across nodes.



Smart Routing (Cost Optimization)

The Hybrid Router intercepts queries. Rule-based or fuzzy matching handles simple tasks (e.g., "How many applicants?" triggers a standard Python count). Zero LLM tokens are wasted.



Specialized Node Activation

If the router detects a complex intent, it hands off to the Gemini 2.5 Flash LLM, which selects the specialized tool/agent node (RAG, Web Search, Interview Gen) needed to execute the step.

Semantic Screening (RAG)

Keyword search is flawed. If a JD requests "AWS Engineer," but a resume says "Built cloud deployment platform," standard filters fail. RAG solves this.

Pipeline: Raw Resumes are vectorized via Sentence Transformers (all-MiniLM-L6-v2) and stored in ChromaDB. Similarity Search then matches candidate experience against the parsed JD, yielding precise match scores and skill-gap analyses.



Human-In-The-Loop (HITL)

The AI Recommends

AI cannot directly hire, reject, or finalize compensation.

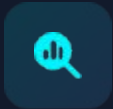
Agent Action: "Based on the JD and ChromaDB screening, I recommend shortlisting Rohan Gupta (94% match) and Anjali Rao (89% match). I have also fetched market salary data via Tavily. Do you approve this shortlist?"

The Human Decides

The recruiter holds the final authority.

Human Action: The recruiter reviews the AI's explanation, confirms the shortlist via the UI, and provides the finalized company budget to proceed to the interview generation node.

Core Agentic Features



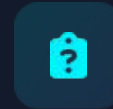
Pydantic JD Parsing

Extracts Role, Skills, Exp, and Location into structured JSON, grounding all subsequent LLM logic.



RAG Candidate Ranking

Screens 16 sample resumes instantly, providing match percentages, missing skills, and detailed explanations.



Interview Gen & Search

Cross-references JD and resume for custom questions. Uses Tavily for live salary data. [Streamlit UI](#)

Criteria & Implementation Mapping

Judging Criteria	HireFlow Implementation
RAG Usage	ChromaDB Vector Screening over 16 text resumes.
Tool Usage	Tavily Live Web Search (Salary) + Python Native Tools (Counting).
Agent Design	Multi-node LangGraph setup with specialized processing agents.
Workflow & Memory	LangGraph state-based routing preserving context and constraints.
Control Systems	Strict Approval System (Human Loop) for shortlisting / hiring limits.

Scaling the Recruitment Co-Pilot



Real PDF Parsing: Transition from text files to robust OCR/PDF parsing for massive scale.



Enterprise ATS Integration: Connecting directly to Workday or Greenhouse APIs.



Autonomous Outreach: Drafting and sending personalized recruiter emails automatically.



"AI assists hiring. Humans make decisions."